The Rolling resistance of a vehicle can be calculated by using the formula

Rolling resistance (R_r) = K.W

Where, W = Weight of the vehicle, Kg

K = Rolling Constant

= 0.005 for good road.

= 0.18 for mud road.

1.6 Gradient resistance

When a vehicle claims on the mountains or high bridges, the complete weight of the vehicle is dragged backwards because of the gravitational force which causes opposing force acting on the vehicle. This opposing force is called as gradient resistance

The gradient Resistance can be calculated by using the following formula

Gradient Resistance (R_G) = W sin θ

Where, W = Weight of vehicle, Kg

θ = Angle between straight and inclined Road

1.7 Tractive effort

When the vehicle is rolling on the road the frictional force between the top of the road and the tyre surface which is in contact on the road is called as the tractive effort.

1.7.1 Traction

Traction is defined as the action which maintains or transfers the tractive effort as same.

1.8 Basic components of Automobile

1. Frame
2. Engine
3. Clutch
4. Gear box
5. Propeller shaft
6. Rear axle
7. Front axle
8. Suspension system
9. Steering system
10. Braking system
11. Wheel and tyre and
12. The electrical system

1.8.1 Important components of the automobile chassis

The vehicle without the body is called as the Chassis. It is the under part of an automobile on which the body is mounted. The Chassis is called as the back bone of the vehicle. The structure of the chassis is shown in the above Figure 1.8.1.

1.8.2 Frame

It is a lengthy component which has cross frames joined rigidly by rivets. It is the foundation for carrying the engine, transmission system, steering system. This frame is made of stainless steel which is able to carry the whole weight of the vehicle. It is back bone of the vehicle.

1.8.3 Engine

The necessary power can be produced from the engine for running

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